DECKING PRODUCT SUBMITTAL

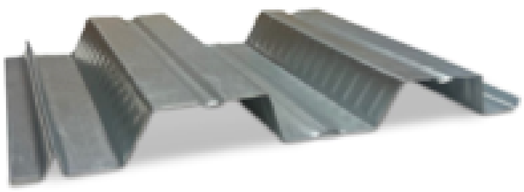
3CD-FY50-18-G90

# DECKING PRODUCT SUBMITTAL

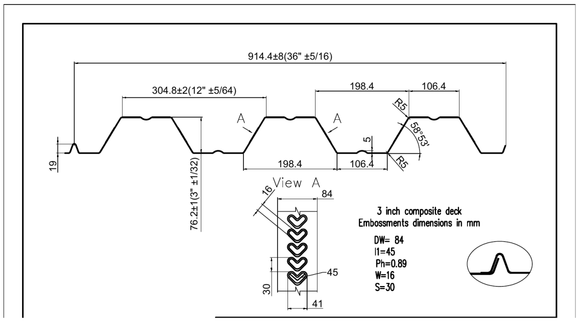
3CD-FY50-18-G90

| PRODUCT | GRADE | GAUGE | COATING |
|---|---|---|---|
| **3" Composite Deck** | **Grade 50** | **18 ga** | **G90** |

## PRODUCT VIEW



## PROFILE DRAWING




## SECTION PROPERTIES · 18 GAUGE

| Thickness (in.) | Weight (psf) | Fy (ksi) | S+ (in³/ft) | S- (in³/ft) | M+ (in-lb/ft) | M- (in-lb/ft) | I+ (in■/ft) | I- (in■/ft) |
|---|---|---|---|---|---|---|---|---|
| 0.0474 | 2.7 | 50 | 0.796 | 0.795 | 23822 | 23812 | 1.274 | 1.260 |

## SOURCE AND USE

1. Source: Refined Metal Steel Catalog 2025, published pages 62, 63, 64, 65, 66
2. ASD section values above are the exact published row for this grade and gauge.
3. Coating selection does not change the published structural performance values.
4. Verify project-specific loading, fastening, span, concrete, and code requirements with the design professional.

REFINED METAL

# PUBLISHED PERFORMANCE DATA

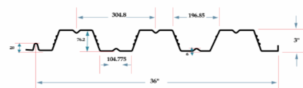
3CD-FY50-18-G90 · 18 GAUGE

## OFFICIAL STEEL CATALOG EXCERPTS

Selected product row is reproduced on Page 1; excerpts retain published table context and notes.

### STEEL CATALOG PAGE 62

GRADE 50 STEEL




**Section Properties**

| Gage | Design Thickness (inches) | Weight (psf) | $F_y$ ksi | $S_e$ + (inch³) per foot | Se - (inch³) per foot | ASD (Ω = 1.67) $M_p$/Ω inch-lbs per ft | ASD (Ω = 1.67) $M_n$/Ω inch-lbs per foot | $I_d$ + (inch⁴) per ft. | $I_d$ - (inch⁴) per ft. |
|---|---|---|---|---|---|---|---|---|---|
| 22 | 0.0295 | 1.7 | 50 | 0.397 | 0.426 | 11895 | 12744 | 0.722 | 0.680 |
| 20 | 0.0358 | 2.1 | 50 | 0.525 | 0.559 | 15709 | 16747 | 0.908 | 0.870 |
| 18 | 0.0474 | 2.7 | 50 | 0.796 | 0.795 | 23822 | 23812 | 1.274 | 1.260 |
| 16 | 0.0598 | 3.5 | 50 | 1.009 | 1.010 | 30200 | 30230 | 1.624 | 1.610 |

**Note**
All section properties and ASD flexural strengths are calculated in accordance with ANSI/SDI RD-2017, AISI S100-2012 and AISI S100-2016.

**Shear and Web Crippling**

| Gage | $V_n$/Ω (lbs/ft) | Web Crippling ($R_n$/Ω), lbs/ft One Flange Loading End Bearing | | | Web Crippling ($R_n$/Ω), lbs/ft One Flange Loading Interior Bearing | | |
|---|---|---|---|---|---|---|---|
| | | 2" | 3" | 4" | 2" | 3" | 4" |
| 22 | 1781 | 402 | 463 | 515 | 643 | 728 | 799 |
| 20 | 2985 | 578 | 663 | 734 | 928 | 1045 | 1143 |
| 18 | 4372 | 974 | 1110 | 1224 | 1575 | 1760 | 1917 |
| 16 | 7370 | 1497 | 1696 | 1864 | 2435 | 2707 | 2936 |

**Note**
All section properties and ASD flexural strengths are calculated in accordance with ANSI/SDI RD-2017, AISI S100-2012 and AISI S100-2016.

**Allowable Uniform Downward Loads, ASD (PSF)**

| Span | Gage | 6' | 7' | 8' | 9' | 10' | 11' | 12' | 13' | 14' | 15' | 16' |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Single | 22 | 220 | 162 | 124 | 98 | 79 | 66 | 55 | 47 | 40 | 35 | 31 |
| | 20 | 291 | 214 | 164 | 129 | 105 | 87 | 73 | 62 | 53 | 47 | 41 |
| | 18 | 441 | 324 | 248 | 196 | 159 | 131 | 110 | 94 | 81 | 71 | 62 |
| | 16 | 559 | 411 | 315 | 249 | 201 | 166 | 140 | 119 | 103 | 89 | 79 |
| Double | 22 | 236 | 173 | 133 | 105 | 85 | 70 | 59 | 50 | 43 | 38 | 33 |
| | 20 | 310 | 228 | 174 | 138 | 112 | 92 | 78 | 66 | 57 | 50 | 44 |
| | 18 | 441 | 324 | 248 | 196 | 159 | 131 | 110 | 94 | 81 | 71 | 62 |
| | 16 | 560 | 411 | 315 | 249 | 202 | 167 | 140 | 119 | 103 | 90 | 79 |
| Triple | 22 | 295 | 217 | 166 | 131 | 106 | 88 | 74 | 63 | 54 | 47 | 41 |
| | 20 | 388 | 285 | 218 | 172 | 140 | 115 | 97 | 83 | 71 | 62 | 55 |
| | 18 | 551 | 405 | 310 | 245 | 198 | 164 | 138 | 117 | 101 | 88 | 78 |
| | 16 | 700 | 514 | 394 | 311 | 252 | 208 | 175 | 149 | 129 | 112 | 98 |

**Notes**
1. All section properties and ASD (Ω = 1.67) uniform loads are calculated in accordance with ANSI/SDI RD-2017, AISI S100-2012 and AISI S100-2016
2. Loads shown in tables are uniformly distributed superimposed loads in psf. Span length assumes center-to-center spacing of supports. Tabulated loads shall not be increased by assuming clear span dimensions.
3. Bending Moment formulae used for flexural stress limitations are: Simple and Two Span $M=\frac{wl^2}{8}$ Three Span or More $M=\frac{wl^2}{10}$
4. Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditi

### STEEL CATALOG PAGE 63

| Span | Gage | 6' | 7' | 8' | 9' | 10' | 11' | 12' | 13' | 14' | 15' | 16' |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Single | 22 | 207 | 130 | 87 | 61 | 45 | 34 | 26 | 20 | 16 | 13 | 11 |
| | 20 | 264 | 167 | 112 | 78 | 57 | 43 | 33 | 26 | 21 | 17 | 14 |
| | 18 | 383 | 241 | 162 | 113 | 83 | 62 | 48 | 38 | 30 | 25 | 20 |
| | 16 | 489 | 308 | 206 | 145 | 106 | 79 | 61 | 48 | 39 | 31 | 26 |
| Double | 22 | 498 | 313 | 210 | 147 | 107 | 81 | 62 | 49 | 39 | 32 | 26 |
| | 20 | 637 | 401 | 269 | 189 | 138 | 103 | 80 | 63 | 50 | 41 | 34 |
| | 18 | 922 | 581 | 389 | 273 | 199 | 150 | 115 | 91 | 73 | 59 | 49 |
| | 16 | 1178 | 742 | 497 | 349 | 254 | 191 | 147 | 116 | 93 | 75 | 62 |
| Triple | 22 | 389 | 245 | 164 | 115 | 84 | 63 | 49 | 38 | 31 | 25 | 21 |
| | 20 | 498 | 314 | 210 | 148 | 108 | 81 | 62 | 49 | 39 | 32 | 26 |
| | 18 | 722 | 454 | 304 | 214 | 156 | 117 | 90 | 71 | 57 | 46 | 38 |
| | 16 | 922 | 581 | 389 | 273 | 199 | 150 | 115 | 91 | 73 | 59 | 49 |

**Note**
For loads that cause L/120 Deflection, multiply by 2.0. For loads that cause L/180 Deflection, multiply by 1.5. For loads that cause L/360 Deflection, multiply by 0.667.

**Construction Span Table – 20 psf Construction Load**

| Normal Weight Concrete (145 pcf) | | | | |
|---|---|---|---|---|
| Total Slab Depth | Deck Type | Maximum Unshored Clear Span 1 span | Maximum Unshored Clear Span 2 span | Maximum Unshored Clear Span 3 span |
| 5.00 (t=2.00) 46 PSF | 3x12x22 ga | 10' 3" | 11' 4" | 11' 9" |
| | 3x12x20 ga | 11' 12" | 13' 0" | 13' 5" |
| | 3x12x18 ga | 13' 5" | 15' 6" | 16' 0" |
| | 3x12x16 ga | 14' 7" | 17' 6" | 18' 1" |
| 5.50 (t=2.50) 52 PSF | 3x12x22 ga | 9' 10" | 10' 10" | 11' 3" |
| | 3x12x20 ga | 11' 6" | 12' 5" | 12' 10" |
| | 3x12x18 ga | 12' 11" | 14' 10" | 15' 4" |
| | 3x12x16 ga | 14' 0" | 16' 9" | 17' 4" |
| 6.00 (t=3.00) 58 PSF | 3x12x22 ga | 9' 5" | 10' 5" | 10' 9" |
| | 3x12x20 ga | 11' 4" | 12' 10" | 13' 4" |
| | 3x12x18 ga | 12' 5" | 14' 3" | 14' 9" |
| | 3x12x16 ga | 13' 6" | 16' 1" | 16' 7" |
| 6.50 (t=3.50) 64 PSF | 3x12x22 ga | 9' 0" | 10' 1" | 10' 5" |
| | 3x12x20 ga | 10' 8" | 11' 6" | 11' 11" |
| | 3x12x18 ga | 12' 0" | 13' 9" | 14' 3" |
| | 3x12x16 ga | 13' 1" | 15' 6" | 16' 0" |
| 7.00 (t=4.00) 70 PSF | 3x12x22 ga | 8' 9" | 9' 9" | 10' 1" |
| | 3x12x20 ga | 10' 3" | 11' 2" | 11' 6" |
| | 3x12x18 ga | 11' 8" | 13' 3" | 13' 9" |
| | 3x12x16 ga | 12' 8" | 15' 0" | 15' 6" |
| 7.50 (t=4.50) 76 PSF | 3x12x22 ga | 8' 5" | 9' 5" | 9' 9" |
| | 3x12x20 ga | 9' 11" | 10' 9" | 11' 2" |
| | 3x12x18 ga | 11' 4" | 12' 10" | 13' 4" |
| | 3x12x16 ga | 12' 4" | 14' 6" | 15' 0" |

| Lightweight Concrete (115 pcf) | | | | |
|---|---|---|---|---|
| Total Slab Depth | Deck Type | Maximum Unshored Clear Span 1 span | Maximum Unshored Clear Span 2 span | Maximum Unshored Clear Span 3 span |
| 5.00 (t=2.00) 37 PSF | 3x12x22 ga | 11' 2" | 12' 3" | 12' 7" |
| | 3x12x20 ga | 12' 11" | 13' 12" | 14' 6" |
| | 3x12x18 ga | 14' 5" | 16' 8" | 17' 3" |
| | 3x12x16 ga | 15' 8" | 18' 10" | 19' 5" |
| 5.50 (t=2.50) 42 PSF | 3x12x22 ga | 10' 8" | 11' 8" | 12' 1" |
| | 3x12x20 ga | 12' 4" | 13' 5" | 13' 10" |
| | 3x12x18 ga | 13' 10" | 16' 0" | 16' 6" |
| | 3x12x16 ga | 15' 0" | 18' 0" | 18' 8" |
| 6.00 (t=3.00) 47 PSF | 3x12x22 ga | 10' 2" | 11' 3" | 11' 8" |
| | 3x12x20 ga | 12' 4" | 14' 2" | 14' 8" |
| | 3x12x18 ga | 13' 4" | 15' 5" | 15' 11" |
| | 3x12x16 ga | 14' 6" | 17' 4" | 17' 11" |
| 6.50 (t=3.50) 49 PSF | 3x12x22 ga | 10' 0" | 11' 1" | 11' 6" |
| | 3x12x20 ga | 11' 9" | 12' 9" | 13' 2" |
| | 3x12x18 ga | 13' 2" | 15' 2" | 15' 8" |
| | 3x12x16 ga | 14' 3" | 17' 1" | 17' 8" |
| 7.00 (t=4.00) 52 PSF | 3x12x22 ga | 9' 10" | 10' 10" | 11' 3" |
| | 3x12x20 ga | 11' 6" | 12' 5" | 12' 10" |
| | 3x12x18 ga | 12' 11" | 14' 10" | 15' 4" |
| | 3x12x16 ga | 14' 0" | 16' 9" | 17' 4" |
| 7.50 (t=4.50) 59 PSF | 3x12x22 ga | 9' 4" | 10' 4" | 10' 9" |
| | 3x12x20 ga | 11' 0" | 11' 11" | 12' 3" |
| | 3x12x18 ga | 12' 4" | 14' 2" | 14' 8" |
| | 3x12x16 ga | 13' 5" | 16' 0" | 16' 6" |

**Note**
Web crippling and shear have not been accounted for in these tables. Required bearing should be determined based on specific span conditions.

REFINED METAL

# PUBLISHED PERFORMANCE DATA

3CD-FY50-18-G90 · 18 GAUGE

## OFFICIAL STEEL CATALOG EXCERPTS

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### STEEL CATALOG PAGE 64

22 ga Normalweight Concrete (145 pcf, f'c = 3,000 psi)

| Slab Thickness (Inches) | Weight (psf) | 8'-0 | 8'-6 | 9'-0 | 9'-6 | 10'-0 | 10'-6 | 11'-0 |
|---|---|---|---|---|---|---|---|---|
| 5 | 46 | 272 | 238 | 210 | 186 | 165 | 147 | 132 |
| 5.5 | 52 | 329 | 288 | 254 | 225 | 200 | 179 | 160 |
| 6 | 58 | 388 | 340 | 300 | 266 | 237 | 212 | 190 |
| 6.5 | 64 | 400 | 394 | 347 | 308 | 274 | 245 | 220 |
| 7 | 70 | 400 | 400 | 396 | 351 | 313 | 280 | 252 |
| 7.5 | 76 | 400 | 400 | 400 | 395 | 352 | 316 | 284 |

| Slab Thickness (Inches) | 11'-6 | 12'-0 | 12'-6 | 13'-0 | 13'-6 | 14'-0 | 14'-6 | 15'-0 |
|---|---|---|---|---|---|---|---|---|
| 5 | 118 | 107 | 96 | 87 | 79 | 71 | 65 | 59 |
| 5.5 | 144 | 130 | 117 | 106 | 96 | 88 | 80 | 72 |
| 6 | 171 | 154 | 140 | 127 | 115 | 105 | 95 | 87 |
| 6.5 | 199 | 179 | 162 | 147 | 134 | 122 | 111 | 102 |
| 7 | 227 | 205 | 186 | 169 | 154 | 140 | 128 | 117 |
| 7.5 | 256 | 231 | 210 | 191 | 174 | 158 | 145 | 132 |

20 ga Normalweight Concrete (145 pcf, f'c = 3,000 psi)

| Slab Thickness (Inches) | Weight (psf) | 8'-0 | 8'-6 | 9'-0 | 9'-6 | 10'-0 | 10'-6 | 11'-0 |
|---|---|---|---|---|---|---|---|---|
| 5 | 46 | 329 | 288 | 254 | 226 | 201 | 180 | 162 |
| 5.5 | 52 | 397 | 349 | 308 | 273 | 244 | 218 | 196 |
| 6 | 58 | 400 | 400 | 363 | 323 | 288 | 258 | 232 |
| 6.5 | 64 | 400 | 400 | 400 | 374 | 334 | 299 | 270 |
| 7 | 70 | 400 | 400 | 400 | 400 | 381 | 342 | 308 |
| 7.5 | 76 | 400 | 400 | 400 | 400 | 400 | 385 | 347 |

| Slab Thickness (Inches) | 11'-6 | 12'-0 | 12'-6 | 13'-0 | 13'-6 | 14'-0 | 14'-6 | 15'-0 |
|---|---|---|---|---|---|---|---|---|
| 5 | 146 | 132 | 119 | 108 | 98 | 90 | 82 | 75 |
| 5.5 | 177 | 160 | 145 | 132 | 120 | 110 | 100 | 92 |
| 6 | 210 | 190 | 172 | 157 | 143 | 131 | 120 | 110 |
| 6.5 | 244 | 221 | 201 | 183 | 167 | 152 | 140 | 128 |
| 7 | 278 | 252 | 229 | 209 | 191 | 175 | 160 | 147 |
| 7.5 | 314 | 285 | 259 | 236 | 216 | 198 | 181 | 166 |

18 ga Normalweight Concrete (145 pcf, f'c = 3,000 psi)

| Slab Thickness (Inches) | Weight (psf) | 8'-0 | 8'-6 | 9'-0 | 9'-6 | 10'-0 | 10'-6 | 11'-0 |
|---|---|---|---|---|---|---|---|---|
| 5 | 46 | 400 | 400 | 381 | 339 | 303 | 273 | 246 |
| 5.5 | 52 | 400 | 400 | 400 | 397 | 355 | 320 | 289 |
| 6 | 58 | 400 | 400 | 400 | 400 | 400 | 370 | 334 |
| 6.5 | 64 | 400 | 400 | 400 | 400 | 400 | 400 | 381 |
| 7 | 70 | 400 | 400 | 400 | 400 | 400 | 400 | 400 |
| 7.5 | 76 | 400 | 400 | 400 | 400 | 400 | 400 | 400 |

| Slab Thickness (Inches) | 11'-6 | 12'-0 | 12'-6 | 13'-0 | 13'-6 | 14'-0 | 14'-6 | 15'-0 |
|---|---|---|---|---|---|---|---|---|
| 5 | 223 | 203 | 185 | 169 | 155 | 142 | 130 | 120 |
| 5.5 | 262 | 238 | 217 | 198 | 182 | 167 | 153 | 141 |
| 6 | 303 | 275 | 251 | 230 | 210 | 193 | 178 | 164 |
| 6.5 | 345 | 314 | 287 | 262 | 241 | 221 | 204 | 188 |
| 7 | 390 | 355 | 324 | 296 | 272 | 250 | 230 | 213 |
| 7.5 | 400 | 397 | 362 | 331 | 304 | 280 | 258 | 238 |

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| (Inches) | (psf) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 5 | 37 | 400 | 400 | 368 | 328 | 294 | 265 | 240 |
| 5.5 | 42 | 400 | 400 | 400 | 385 | 345 | 310 | 281 |
| 6 | 47 | 400 | 400 | 400 | 400 | 399 | 359 | 325 |
| 6.5 | 49 | 400 | 400 | 400 | 400 | 400 | 400 | 373 |
| 7 | 52 | 400 | 400 | 400 | 400 | 400 | 400 | 400 |
| 7.5 | 59 | 400 | 400 | 400 | 400 | 400 | 400 | 400 |

| Slab Thickness (Inches) | 11'-6 | 12'-0 | 12'-6 | 13'-0 | 13'-6 | 14'-0 | 14'-6 | 15'-0 |
|---|---|---|---|---|---|---|---|---|
| 5 | 217 | 198 | 181 | 166 | 152 | 140 | 129 | 119 |
| 5.5 | 255 | 232 | 212 | 194 | 178 | 164 | 151 | 140 |
| 6 | 295 | 269 | 245 | 225 | 207 | 190 | 176 | 162 |
| 6.5 | 339 | 309 | 282 | 259 | 238 | 219 | 203 | 188 |
| 7 | 384 | 350 | 320 | 294 | 270 | 249 | 230 | 213 |
| 7.5 | 400 | 390 | 357 | 327 | 301 | 278 | 257 | 237 |

16 ga Lightweight Concrete (115 pcf, f'c = 3,000 psi)

| Slab Thickness (Inches) | Weight (psf) | 8'-0 | 8'-6 | 9'-0 | 9'-6 | 10'-0 | 10'-6 | 11'-0 |
|---|---|---|---|---|---|---|---|---|
| 5 | 37 | 400 | 400 | 391 | 349 | 313 | 282 | 255 |
| 5.5 | 42 | 400 | 400 | 400 | 400 | 380 | 343 | 310 |
| 6 | 47 | 400 | 400 | 400 | 400 | 400 | 400 | 369 |
| 6.5 | 49 | 400 | 400 | 400 | 400 | 400 | 400 | 400 |
| 7 | 52 | 400 | 400 | 400 | 400 | 400 | 400 | 400 |
| 7.5 | 59 | 400 | 400 | 400 | 400 | 400 | 400 | 400 |

| Slab Thickness (Inches) | 11'-6 | 12'-0 | 12'-6 | 13'-0 | 13'-6 | 14'-0 | 14'-6 | 15'-0 |
|---|---|---|---|---|---|---|---|---|
| 5 | 231 | 211 | 193 | 177 | 162 | 149 | 138 | 127 |
| 5.5 | 282 | 257 | 235 | 215 | 198 | 182 | 168 | 156 |
| 6 | 335 | 306 | 280 | 257 | 236 | 217 | 201 | 186 |
| 6.5 | 393 | 358 | 328 | 301 | 277 | 256 | 237 | 219 |
| 7 | 400 | 400 | 378 | 347 | 320 | 295 | 273 | 253 |
| 7.5 | 400 | 400 | 400 | 391 | 361 | 333 | 308 | 286 |